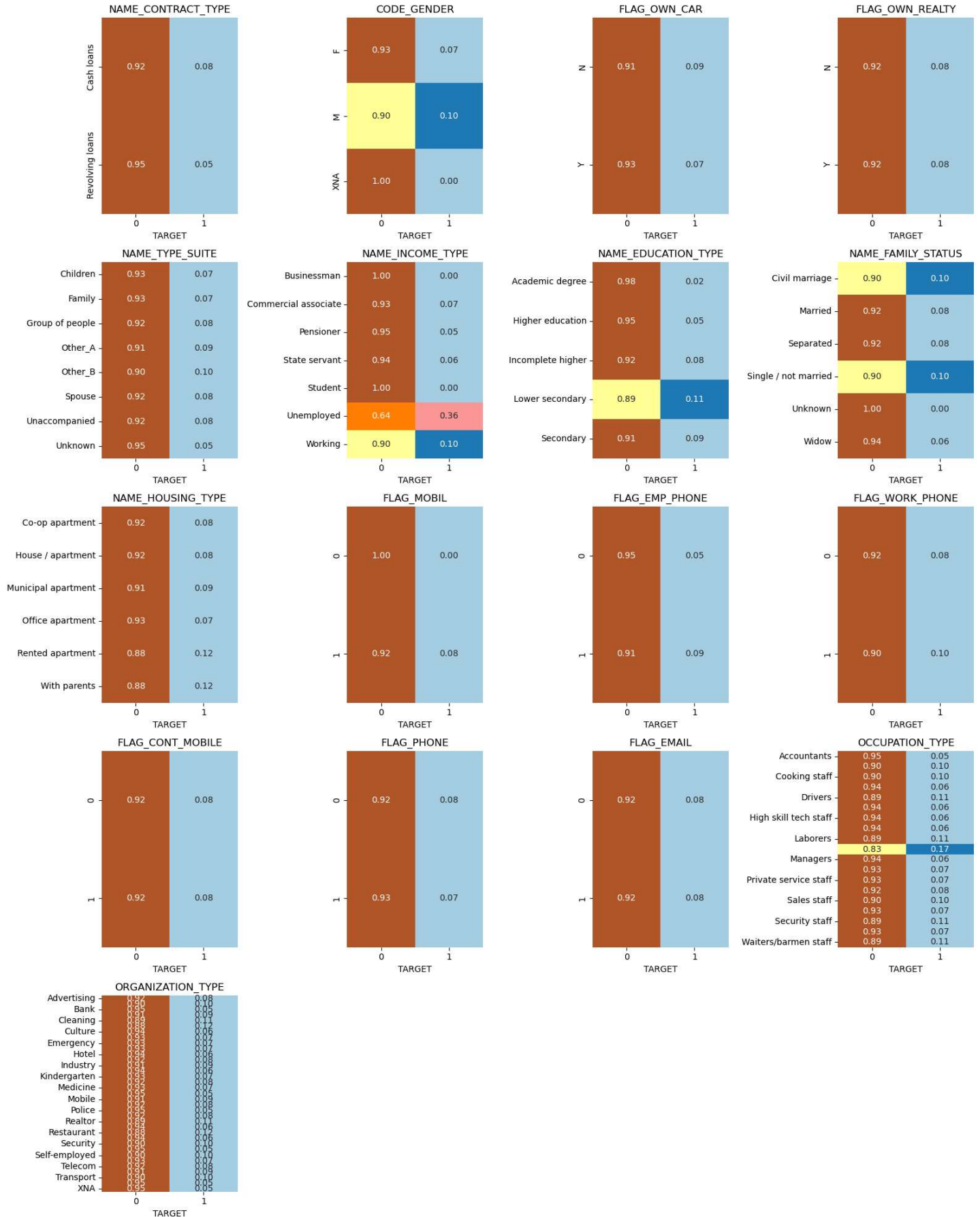
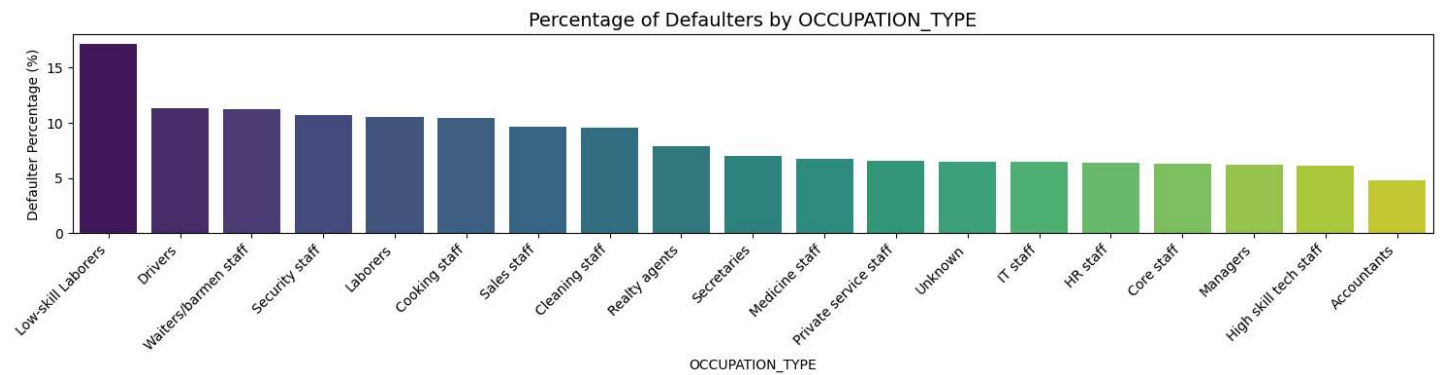
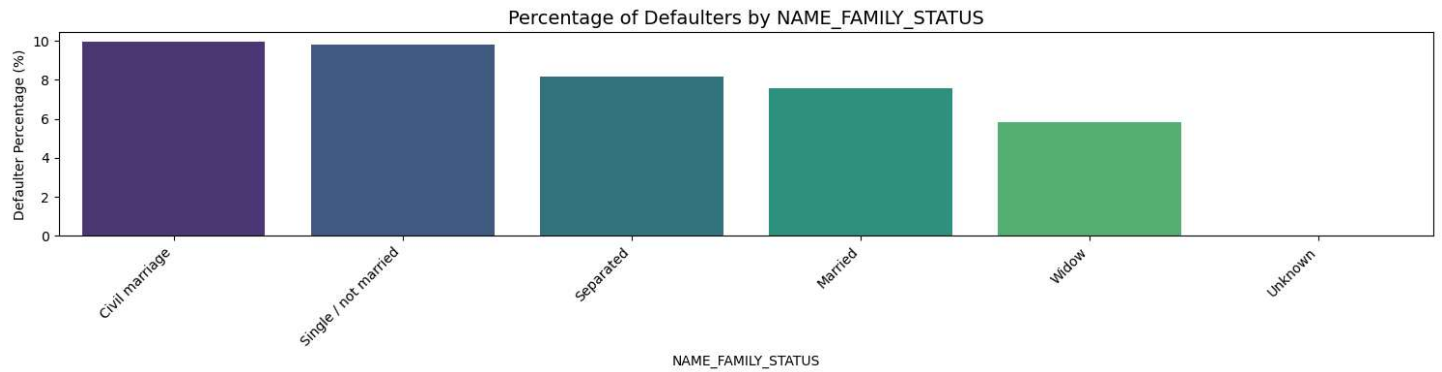
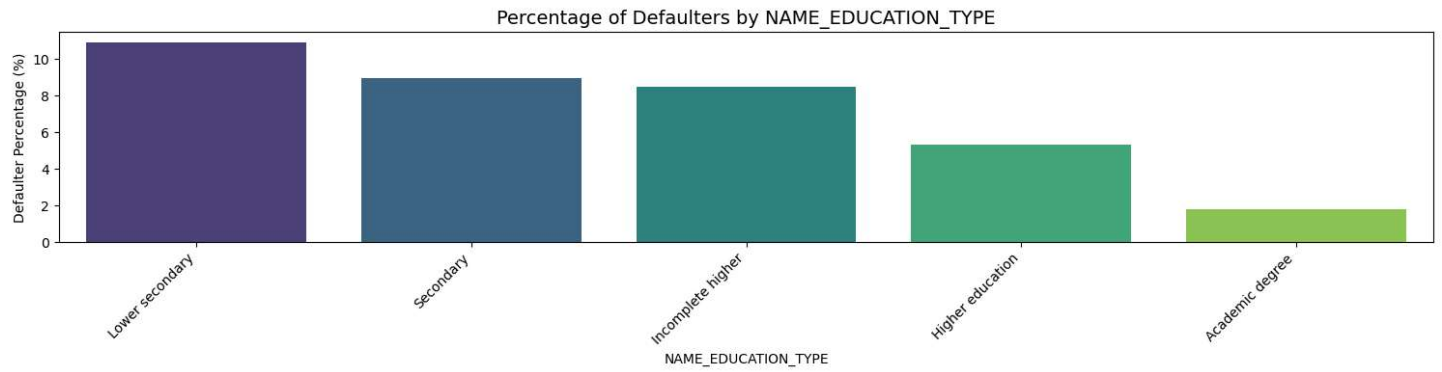
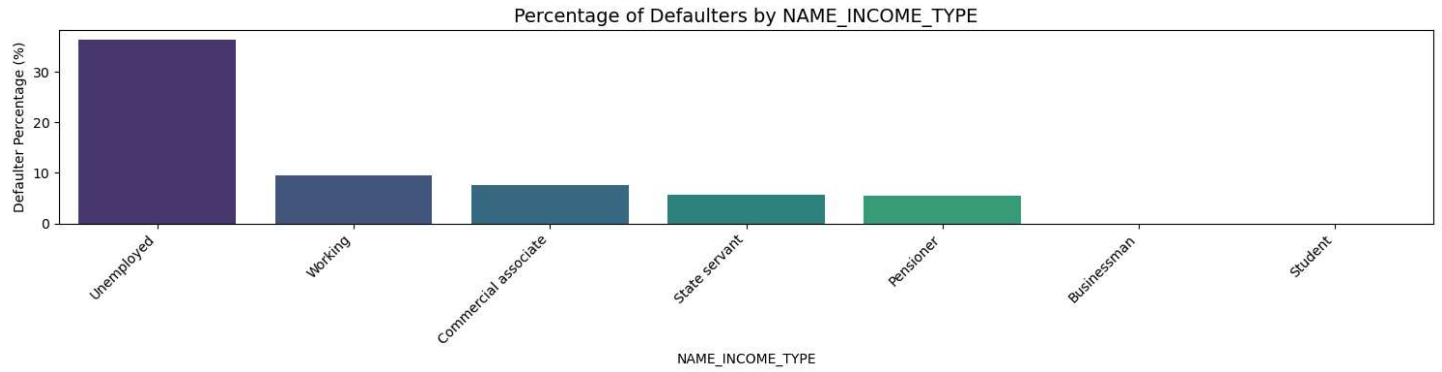
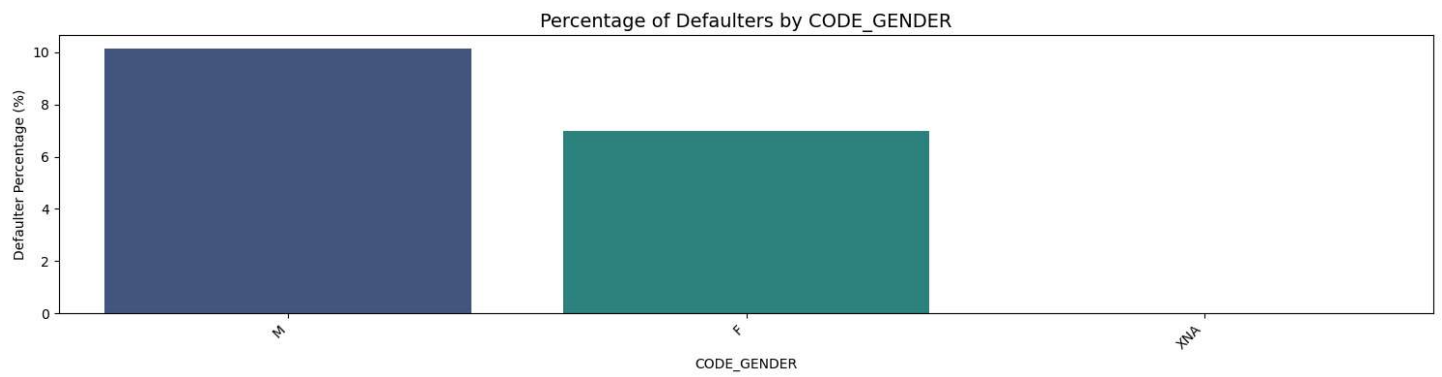


EDA On Application Data

Relation between Categorical Variables and Default (TARGET)

Crosstabulation of Categorical Features vs. TARGET (Defaulter %)





Key Insights from Categorical Features vs TARGET (Defaulters %)

1. Employment Status:

- **Unemployed (36%)** show the **highest default rate** among all categories.
- **Businessman (0%)** and **Students (0%)** have the **lowest risk** of default.

2. Education Level:

- Clients with **Lower Secondary Education (11%)** have a **higher chance of default**.
- Those with an **Academic Degree (2%)** are the most reliable.

3. Gender:

- **Males (10%)** default more than **Females (7%)**.

4. Loan Type:

- **Cash Loans (8%)** have a higher default rate than **Revolving Loans (5%)**.

5. Housing Type:

- People living **Rented (12%)** or **With Parents (12%)** show higher defaults.
- **House/Apartment owners (8%)** are more stable borrowers.

6. Family Status:

- **Single or Civil Marriage clients (10%)** have higher default rates than **Married (8%)**.

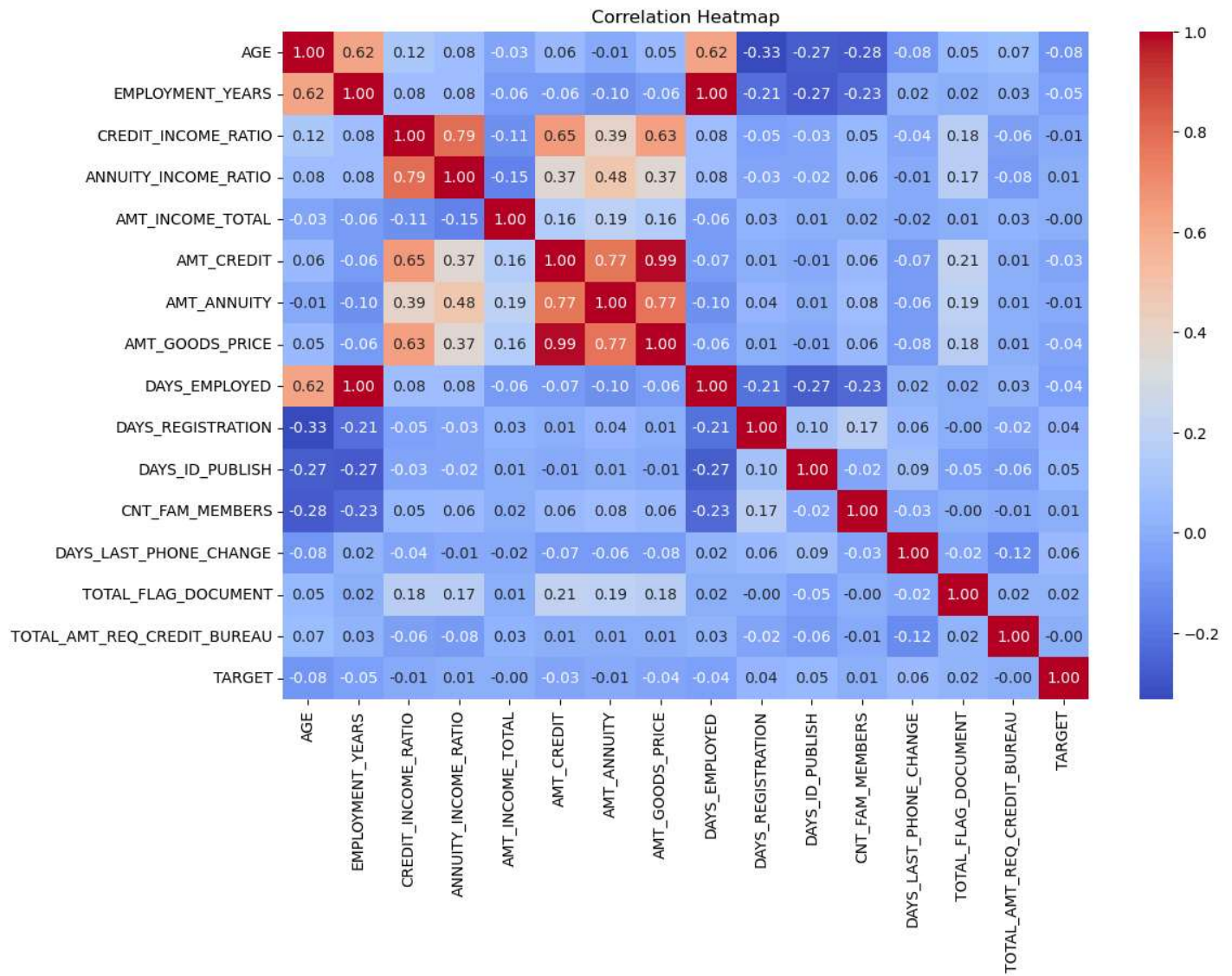
7. Occupation Type:

- **Low-skill Laborers (17%)** show the **highest default** among occupations.
- **Accountants (5%)** and **High-skill tech staff (6%)** are more creditworthy.

8. General Observation:

- Most categorical groups cluster around **7–9% default rate**.
- Employment, Education, income type, and housing situation are the **strongest predictors** of loan default likelihood.

Relation between Numerical Variables and Default (TARGET)

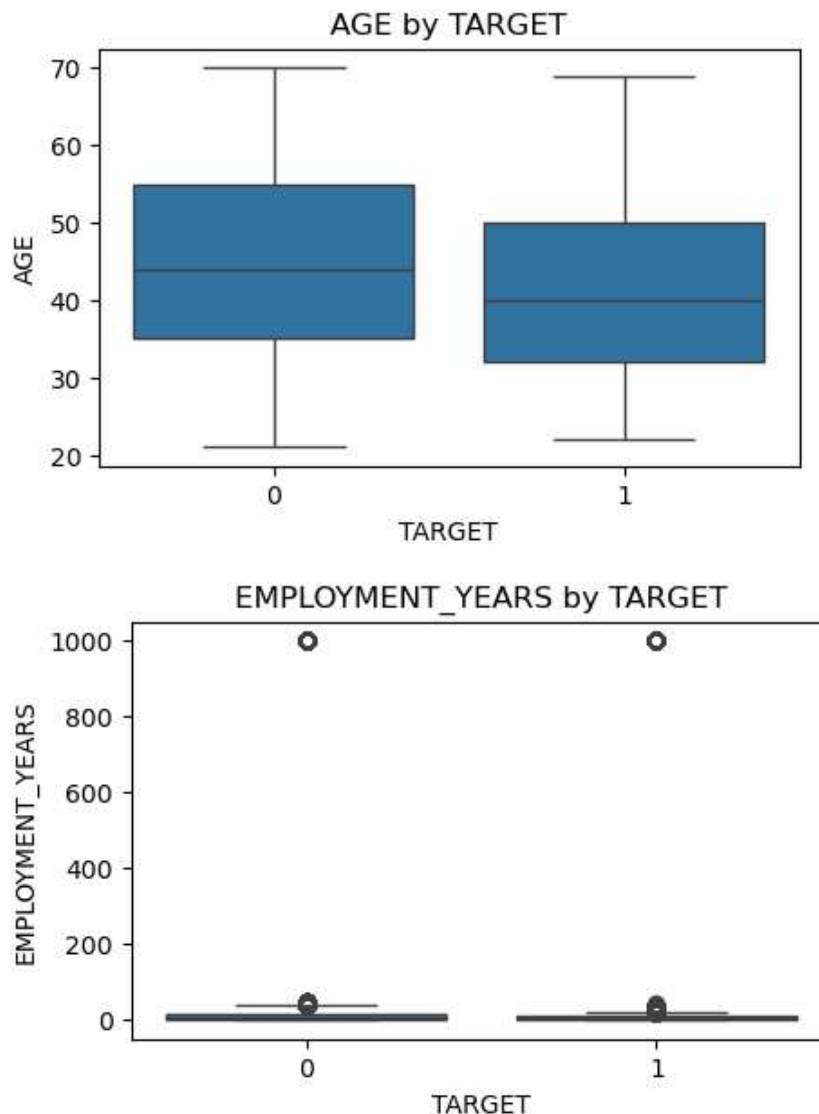


Correlation of Numerical Features with TARGET (Default)

Feature	Correlation with TARGET
AGE	-0.08
EMPLOYMENT_YEARS	-0.05
CREDIT_INCOME_RATIO	0.01
ANNUITY_INCOME_RATIO	0.01
AMT_INCOME_TOTAL	-0.00
AMT_CREDIT	-0.03
AMT_ANNUITY	-0.04
AMT_GOODS_PRICE	-0.03
DAYS_EMPLOYED	-0.04
DAYS_REGISTRATION	0.04
DAYS_ID_PUBLISH	0.05
CNT_FAM_MEMBERS	0.01
DAYS_LAST_PHONE_CHANGE	0.06

Key Observations

- All correlations with `TARGET` are **very weak** ($|r| < 0.1$) → no single numeric variable strongly predicts default.
- **Negative correlations:**
 - `AGE` (-0.08) and `EMPLOYMENT_YEARS` (-0.05) → Younger or less experienced clients tend to default slightly more.
- **Positive correlations:**
 - `DAYS_LAST_PHONE_CHANGE` (0.06) and `DAYS_ID_PUBLISH` (0.05) → More recent updates or changes in personal info slightly increase default risk.
- **New ratio features** (`CREDIT_INCOME_RATIO` , `ANNUITY_INCOME_RATIO`) show almost zero correlation with `TARGET` , but they may help in **non-linear models** (like tree-based algorithms).

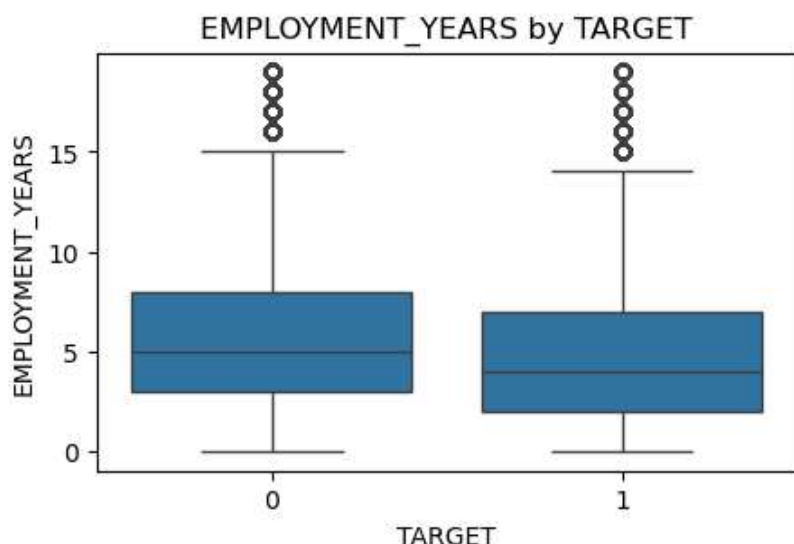


Key Observations (Derived from Box Plots: Defaulters vs. Non-Defaulters)

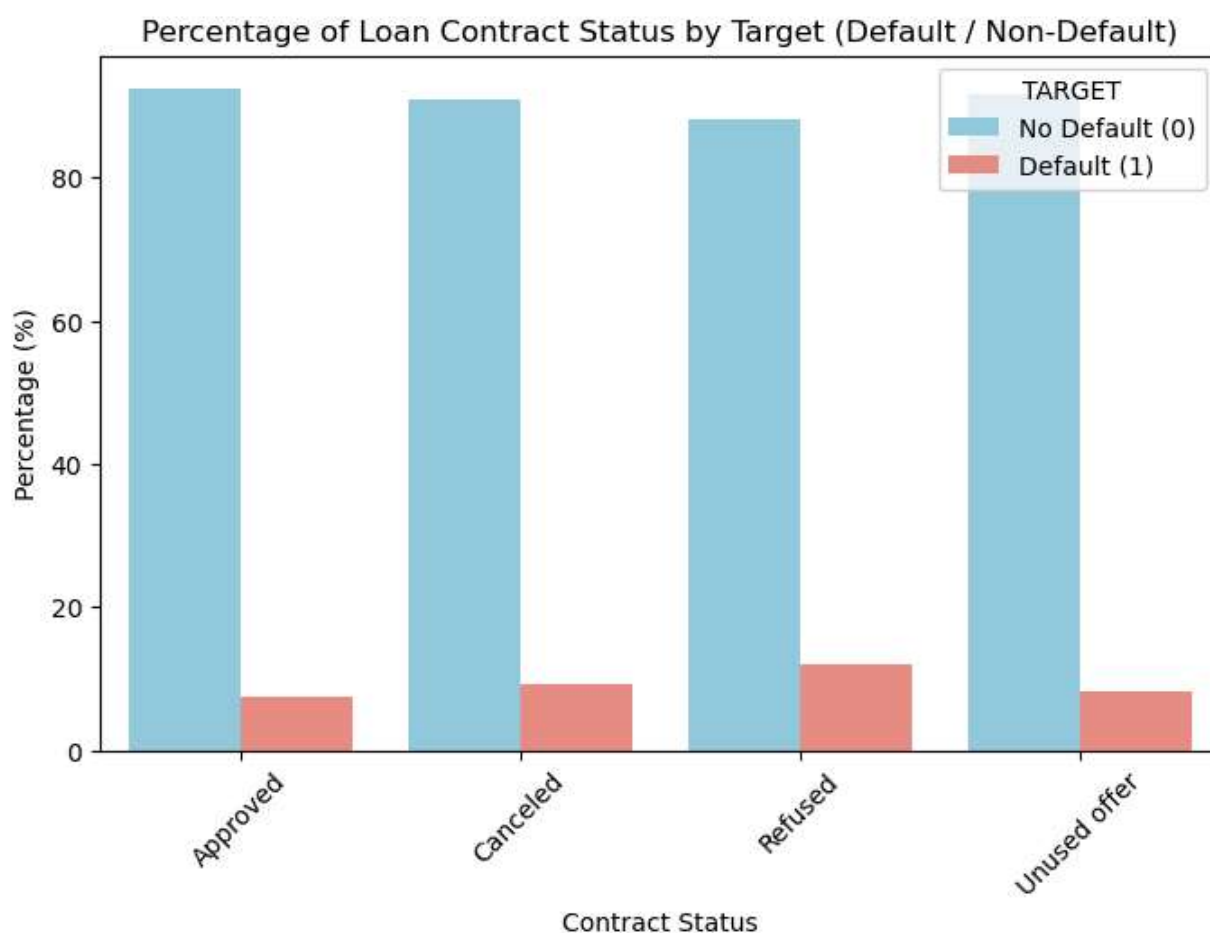
- **Differences in central tendency (Median/IQR) are minor**, suggesting that neither AGE nor EMPLOYMENT_YEARS alone strongly distinguishes defaulters from non-defaulters.
- **Defaulters are Younger (AGE):**
 - **Non-Defaulters (`TARGET=0`):** Have a slightly higher median age (around 45 years).
 - **Defaulters (`TARGET=1`):** Have a slightly lower median age (around 40 years).
 - **Insight:** *Non-defaulters tend to be slightly older than defaulters.*

Crucial Data Quality Insight (EMPLOYMENT_YEARS)

- **Extreme Outliers in EMPLOYMENT_YEARS** : Both defaulters and non-defaulters show extreme outliers reaching up to 1000 years.



EDA On Previous Application Data



Key Observations (Contract Status by Default Status) - Previous Application Data

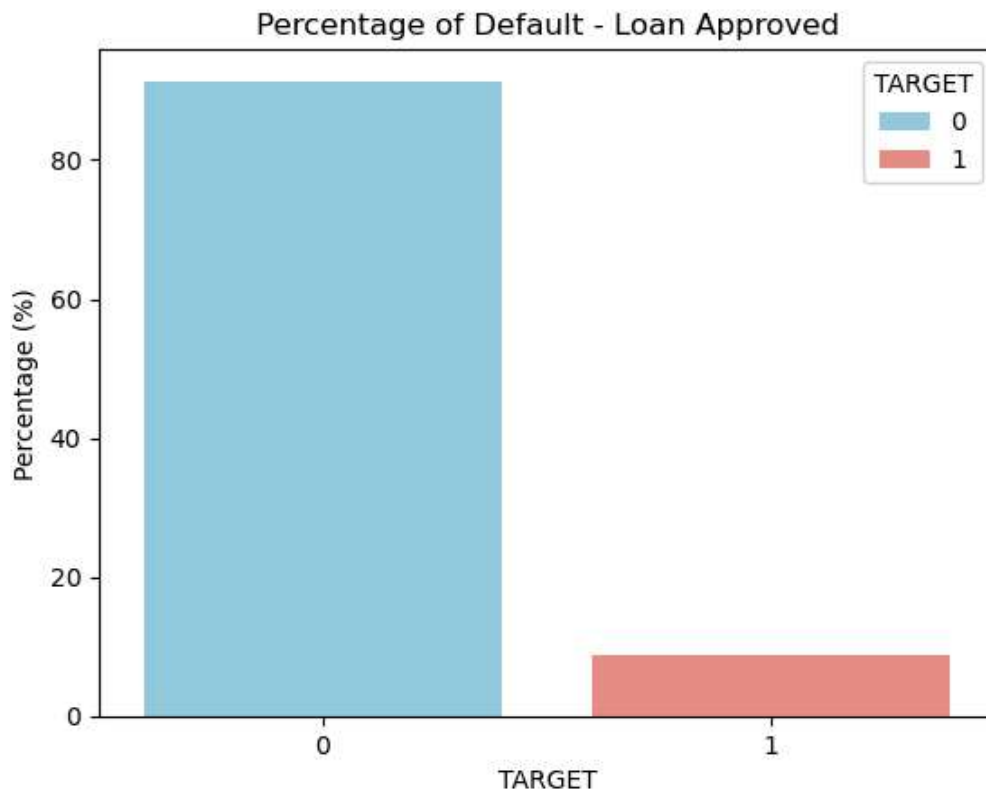
- **"Refused" Loans Have the Highest Default Rate:**
 - Among all the categories shown, the **"Refused"** contract status has the highest percentage of **Defaulters (1)**, at approximately 12-13%.
- **"Approved" and "Unused offer" Have the Lowest Default Rates:**

- The default rate is lowest for clients whose previous contract was "**Approved**" or "**Unused offer**," both sitting around 7-8%.
- "**Canceled**" Status is Mid-Range:
 - The default rate for clients with a "**Canceled**" previous contract is intermediate, around 9-10%.

Approved Loan - percentage of Defaulters - 7.6%

Out[25]:

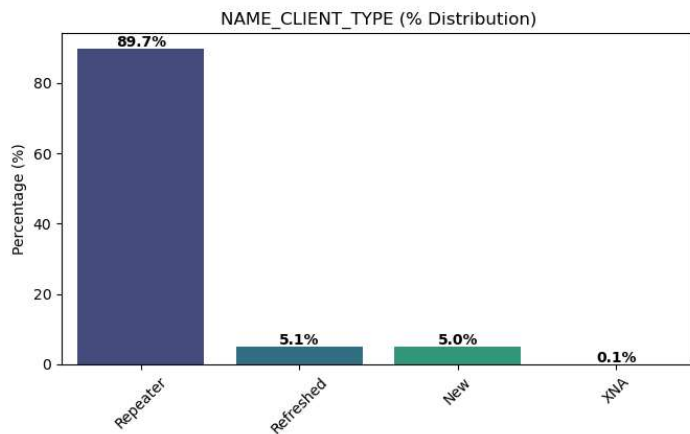
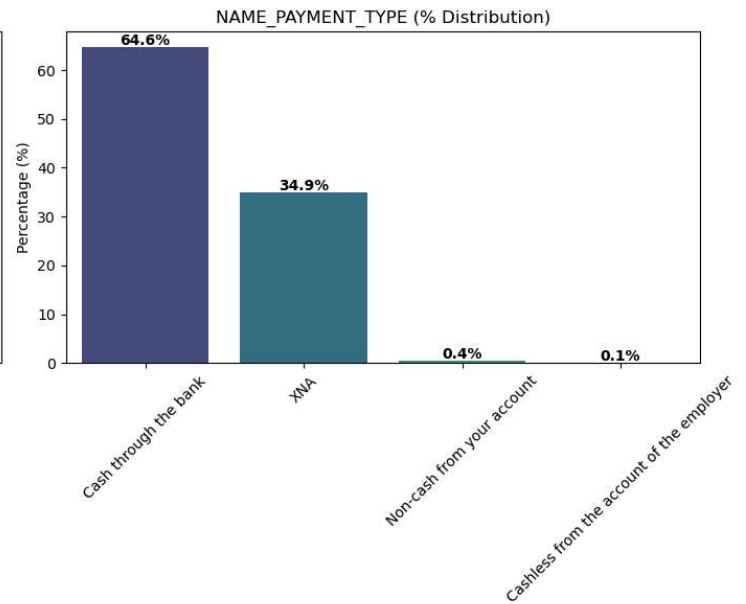
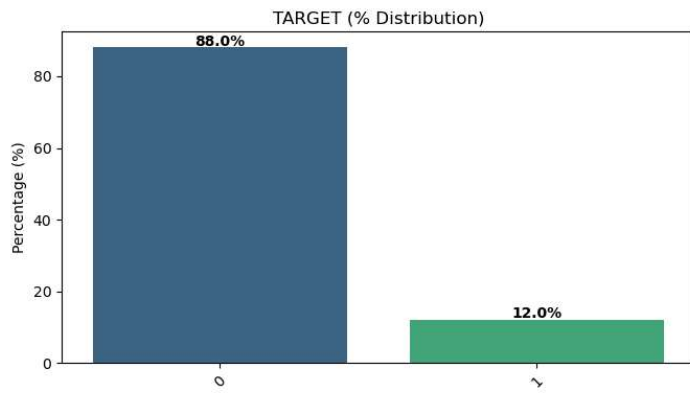
	TARGET	Percentage
0	0	92.411345
1	1	7.588655



Key Observations (Percentage of Default - Loan Approved) - Previous Application Data

- "For Approved Loan"
 - **Defaulters (TARGET=1) are approximately 7.6%** of the clients whose *previous* loan application status was "Approved."
 - **Insight:** The 7.6% represents the realized financial loss (defaulters) among the previous loan applications that were approved. *This is the direct credit risk the bank incurred for that group, which could theoretically have been avoided if these specific defaulters were identified and denied the loan.*

Refused Loan - percentage of Defaulters - 12%



Key Observations (Percentage of Default - Loan Refused) - Previous Application Data

- **"For Refused Loan"**
 - Defaulters (TARGET=1) are approximately 12% of the clients whose *previous* loan application status was "Refused."
 -
- **"Refused" Status:**
 - **Higher Refusal Rate:** Clients using the payment type "**cash through the bank**" and those identified as "**Repeater**" clients have historically experienced a higher proportion of loan refusals.
 - **Lower Refusal Rate:** Clients utilizing the payment type "**cashless from the account of the employer**" (suggesting direct salary deposit/stability) have historically experienced fewer loan refusals.